

Requirement Analysis Document

Project: TaskFlow - To-Do Application
Document Version: 1.0.0
Date: April 6, 2026
Author: Requirements Analysis Team
Status: Final

Document Control

Version	Date	Author	Changes
1.0.0	2026-04-06	Requirements Analysis Team	Initial comprehensive RA document creation

1. Introduction

1.1 Project Overview

TaskFlow is a cross-platform task management application designed to enhance personal productivity through intuitive task organization, tracking, and prioritization. The application provides a unified experience across web and mobile platforms, enabling users to seamlessly manage daily

activities with features including task creation, categorization, status tracking, priority management, and intelligent notifications. TaskFlow adopts a minimalist design philosophy to reduce cognitive load and maximize user focus on task completion.

1.2 Purpose

This Requirement Analysis (RA) Document translates the Software Requirements Specification (SRS) into a detailed, actionable framework for development teams, architects, and project stakeholders. It provides granular elaboration of functional and non-functional requirements, defines data architecture, identifies constraints, and recommends technology solutions to guide the design, development, and validation phases of the TaskFlow project.

1.3 Scope

The TaskFlow application encompasses: - **Core Functionality:** User authentication, task lifecycle management (CRUD operations), categorization, priority assignment, and status tracking - **Platform Support:** Responsive web application and native mobile applications (iOS/Android) - **Notification System:** Multi-channel reminder system (push notifications and email) - **Data Persistence:** Secure cloud-based storage with real-time synchronization across devices - **User Experience:** Minimalist, accessibility-compliant interface with theme customization

Out of Scope for Initial Release: - Collaborative task sharing between multiple users - Third-party calendar integrations - AI-powered task suggestions - Advanced reporting and analytics dashboards

1.4 Objectives

- Deliver a production-ready MVP within 4-6 months
- Achieve 99.9% system uptime with sub-500ms synchronization latency
- Support 10,000+ concurrent active users
- Maintain <2-second page load times across all supported devices
- Pass security audit compliance (OWASP Top 10 mitigation)

1.5 Target Audience

- **Primary Users:** Professionals and students aged 18-45 seeking personal productivity tools
- **Secondary Users:** Freelancers and remote workers requiring cross-device task synchronization
- **Administrative Users:** System administrators for platform maintenance and monitoring

2. SRS Summary

Metric	Value	Details
Total Functional Requirements	8	Core user-facing features from authentication to notifications
Total Non-Functional Requirements	8	Performance, usability, reliability, and security constraints
Complexity Estimate	Medium	Requires real-time sync, multi-platform support, and secure authentication
Primary User Roles	1	End User (with full CRUD permissions)
Core Data Entities	4	User, Task, Tag, Notification

Platform Targets	3	Web (responsive), iOS, Android
Estimated Development Timeline	4-6 months	For MVP including testing and deployment

Key Features Identified: - Secure JWT-based authentication with password reset workflow - Flexible task management with tagging and priority systems - Real-time cross-device synchronization - Proactive notification system with 30-minute advance reminders - Minimalist, responsive UI with dark/light theme support

3. Functional Requirements

FR-1: User Authentication & Authorization

Description: The system shall provide secure user account management capabilities including registration, authentication, and password recovery. Users must create accounts using a valid email address and password. Upon successful authentication, the system shall issue a JWT token for session management.

Detailed Sub-Requirements: 1.1. **User Registration:** New users can sign up by providing a unique email address, strong password (minimum 8 characters, including uppercase, lowercase, number, and special character), and optional full name. The system must verify email uniqueness and send a confirmation email.

1.2. **User Login:** Registered users can authenticate using their email and password. The system shall implement rate limiting (5 attempts per 15 minutes) to prevent brute-force attacks.

1.3. **Password Reset:** Users can request a password reset link via email. The link must expire after 1 hour and be single-use. Temporary tokens should be stored securely with expiration timestamps.

1.4. **Session Management:** JWT tokens shall be valid for 24 hours with optional "remember me" functionality extending to 7 days. Tokens must include claims for user ID and email.

1.5. **Logout:** Users can explicitly logout, which invalidates the JWT token on the client side.

Acceptance Criteria: - Account creation completes in <3 seconds - Password reset email delivered within 60 seconds - JWT tokens are properly signed and validated on each API request - Rate limiting blocks IP after 5 failed attempts

FR-2: Task Creation

Description: Authenticated users shall be able to create new tasks with mandatory title and optional description, due date, priority, and tags. Tasks are private to the creating user by default.

Detailed Sub-Requirements: 2.1. **Task Title:** Required field, maximum 200 characters, cannot be empty or whitespace-only. 2.2. **Task Description:** Optional field, maximum 2000 characters, supporting rich text formatting (bold, italics, bullet points). 2.3. **Due Date:** Optional datetime field must allow future dates only (minimum 1 minute from creation time). 2.4. **Quick Creation:** Users can create tasks with title-only via a "quick add" input field. 2.5. **Validation:** All task creation attempts must validate input fields and return user-friendly error messages.

Acceptance Criteria: - Task appears in user's list within 500ms of creation - Invalid due dates (past dates) are rejected with clear error messaging - Rich text description renders correctly across all platforms - Quick-add feature supports keyboard shortcut (e.g., Ctrl+N)

FR-3: Task Categorization

Description: Users shall be able to organize tasks using a flexible tagging system. Tags act as labels that can be applied to multiple tasks for filtering and visual grouping.

Detailed Sub-Requirements: 3.1. **Tag Creation:** Users can create custom tags with a name (max 50 characters) and optional color (hex code). 3.2. **Tag Management:** Tags can be edited, deleted, or archived. Deleting a tag removes it from all associated tasks. 3.3. **Multiple Tags:** A single task

can have unlimited tags applied. 3.4. **Predefined Tags:** System provides default tags: "Work", "Personal", "Urgent", "Important". 3.5. **Tag Suggestions:** Auto-complete tag names during task creation based on user's existing tags.

Acceptance Criteria: - Tag changes reflect on all associated tasks within 1 second - Tag names are case-insensitive unique per user - Color picker provides accessible color options with contrast ratios >4.5:1

FR-4: Status Management

Description: Users shall be able to track task progression through a defined status workflow: Pending → In Progress → Completed.

Detailed Sub-Requirements: 4.1. **Status Values:** Three mutually exclusive statuses: - **Pending:** Default status for newly created tasks - **In Progress:** Indicates active work - **Completed:** Task is finished (moves to archived/completed view)

4.2. **Status Transition:** Users can change status via dropdown, drag-and-drop, or swipe gestures (mobile). 4.3. **Completion Timestamp:** System records the datetime when a task is marked as Completed. 4.4. **Bulk Operations:** Users can select multiple tasks and change status in batch.

Acceptance Criteria: - Status changes persist across device sync within 500ms - Completed tasks are filtered from default "Active" view - Completion date is displayable in task details - Bulk operations support selection of up to 50 tasks

FR-5: Priority Levels

Description: Users shall be able to assign priority levels to tasks to enable effective triage and focus on high-impact activities.

Detailed Sub-Requirements: 5.1. **Priority Values:** Three levels with visual indicators: - **Low:** Green indicator, default for tasks without explicit priority - **Medium:** Yellow indicator - **High:** Red indicator with subtle pulse animation

5.2. **Priority Filtering:** Tasks can be sorted and filtered by priority level. 5.3. **Visual Cues:** Priority indicators appear on all task cards and in list views. 5.4. **Smart Defaults:** Tasks with "Urgent" tag and due date <24 hours auto-suggest High priority.

Acceptance Criteria: - Priority filters return results in <1 second - Visual indicators are accessible (screen reader compatible) - Priority changes trigger re-sorting in real-time

FR-6: Task Deletion

Description: Users shall be able to permanently delete individual tasks or perform bulk deletion of completed tasks to maintain data hygiene.

Detailed Sub-Requirements: 6.1. **Single Task Deletion:** Users can delete a task with a confirmation dialog to prevent accidental loss. 6.2. **Bulk Clear Completed:** One-click operation to delete all tasks marked as Completed older than 30 days. 6.3. **Soft Delete Option:** System should support "archive" functionality as alternative to permanent deletion (configurable in settings). 6.4. **Undo Capability:** Deleted tasks can be recovered within 10 seconds via an "Undo" toast notification.

Acceptance Criteria: - Deletion confirmation dialog requires explicit user action - Bulk operation processes 1000+ completed tasks in <5 seconds - Undo functionality successfully restores task with all metadata intact

FR-7: Search & Filter

Description: Users shall be able to locate tasks efficiently through keyword search and multi-criteria filtering across all task metadata.

Detailed Sub-Requirements: 7.1. **Keyword Search:** Full-text search across task title and description with fuzzy matching (typo tolerance). 7.2.

Filter Criteria: Users can combine multiple filters: - Date range (due date) - Priority level - Status - Tags (include/exclude) - Creation date

7.3. **Saved Searches:** Users can save frequently used filter combinations as "Smart Lists". 7.4. **Search Performance:** Search must be debounced (300ms) to optimize performance. 7.5. **Search History:** Last 10 searches stored locally for quick access.

Acceptance Criteria: - Search returns results from 10,000 tasks in <1 second - Fuzzy matching handles up to 2 character typos - Combined filters accurately narrow results - Saved searches sync across all user devices

FR-8: Reminder Notifications

Description: The system shall automatically send proactive notifications to users 30 minutes before a task's due date/time via push and/or email channels.

Detailed Sub-Requirements: 8.1. **Notification Timing:** Reminders trigger exactly 30 minutes before due datetime (configurable 15-60 min range in settings). 8.2. **Delivery Channels:** - **Push Notifications:** Via Firebase Cloud Messaging (Android) and APNs (iOS) - **Email Notifications:** HTML formatted emails with task details and direct links 8.3. **Notification Preferences:** Users can enable/disable each channel globally or per-tag. 8.4. **Snooze Function:** Users can snooze reminders for 15, 30, or 60 minutes. 8.5. **Failure Handling:** Failed notifications retry up to 3 times with exponential backoff.

Acceptance Criteria: - Notifications delivered within ± 1 minute of scheduled time - Push notification delivery rate >98% - Email delivery rate >99% - Users receive no more than 1 reminder per task

4. Non-Functional Requirements

4.1 Performance Requirements

NFR-1: Task List Load Time

The application must load and display the initial task list (up to 100 tasks) in under 2 seconds on a standard 4G connection (≥ 10 Mbps) for authenticated users.

Measurement: Time from API request initiation to render completion.

Acceptance Criteria: 95th percentile load time <1.5s; 99th percentile <2s.

NFR-2: Real-Time Synchronization Latency

Changes made on one device must reflect on all other user devices within 500 milliseconds under normal network conditions.

Measurement: Round-trip time from client A update → server → client B update.

Acceptance Criteria: Median latency <300ms; P99 latency <500ms.

Technology Implication: Requires WebSocket or Server-Sent Events implementation.

4.2 Usability Requirements

NFR-3: Minimalist Design Compliance

The user interface must adhere to minimalist design principles: maximum 5 primary UI colors, ≤2 font families, no more than 3 actions per screen, and generous whitespace usage.

Measurement: Design review against minimalist heuristic checklist.

Acceptance Criteria: Score ≥90/100 on minimalist design audit; user cognitive load test score <3.5/5.

NFR-4: Cross-Device Responsiveness

The application must provide a fully functional, optimized experience across mobile (320px-768px), tablet (768px-1024px), and desktop (>1024px) viewports without horizontal scrolling.

Measurement: BrowserStack testing across 20+ device configurations.

Acceptance Criteria: 100% functional parity across breakpoints; Lighthouse mobile score >90.

4.3 Reliability Requirements

NFR-5: Data Persistence & Durability

All task data must be persisted to a permanent storage layer with zero data loss on page refresh, browser restart, or device change. Database must

support point-in-time recovery within last 7 days.

Measurement: RPO (Recovery Point Objective) and RTO (Recovery Time Objective) testing.

Acceptance Criteria: RPO <5 minutes; RTO <30 minutes; 100% data consistency across sessions.

NFR-6: System Availability

The system must maintain 99.9% uptime (maximum 43.8 minutes downtime/month), excluding scheduled maintenance windows.

Measurement: Uptime monitoring via synthetic transactions every 30 seconds.

Acceptance Criteria: 99.9% availability over 90-day rolling window; scheduled maintenance announced 48 hours in advance.

4.4 Security Requirements

NFR-7: Password Hashing

All user passwords must be hashed using bcrypt algorithm with a cost factor of 12 or higher before storage in the database. Plain text passwords must never be logged or stored.

Measurement: Security audit of authentication module codebase.

Acceptance Criteria: Pass OWASP ASVS Level 1 requirements; bcrypt implementation verified via penetration testing.

NFR-8: API Endpoint Protection

All RESTful API endpoints (except `/auth/login` and `/auth/register`) must validate and verify JWT tokens on each request. Tokens must include expiration, issuer, and audience claims.

Measurement: Automated security scanning of all endpoints.

Acceptance Criteria: 100% endpoint coverage; unauthorized access attempts return 401/403 status; no JWT secrets exposed in client-side code.

5. User Roles & Stakeholders

5.1 Primary User Roles

Role	Description	Permissions & Capabilities
End User	Registered individual managing personal tasks	- Full CRUD on own tasks - Manage personal tags - Configure notification preferences - View own dashboard only - Export personal data

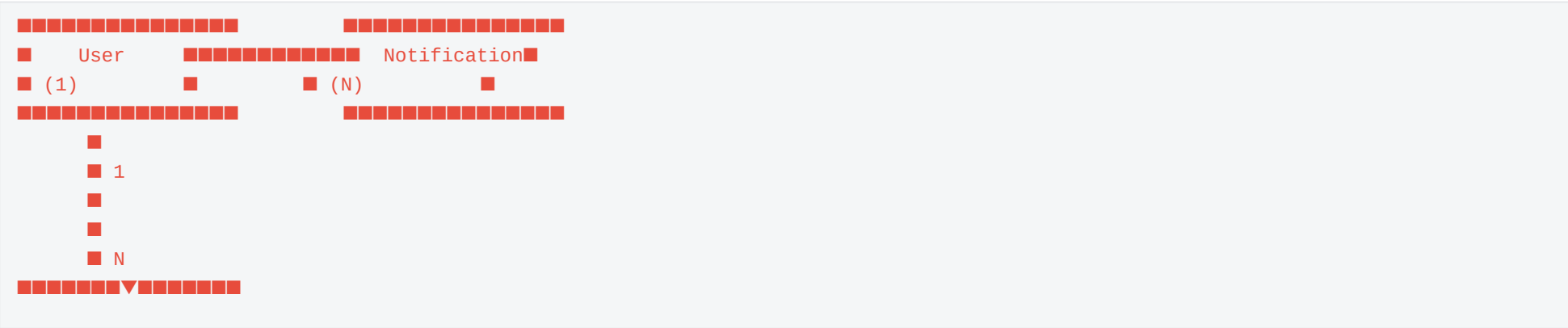
5.2 Stakeholder Identification

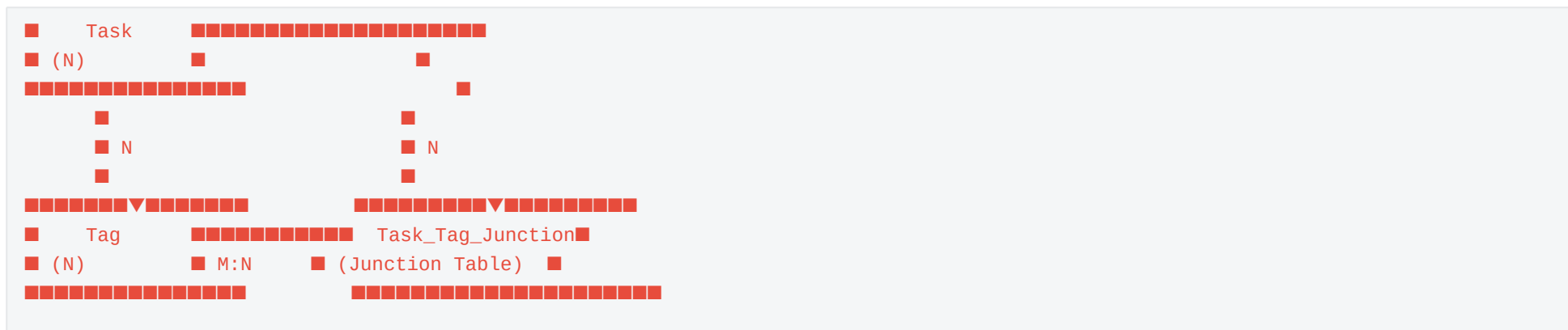
Stakeholder	Interest/Responsibility	Success Criteria
Product Sponsor	Funding, ROI, market positioning	On-time MVP launch; user adoption >5K in first month
Development Team	Technical implementation, code quality	Clean architecture; <5% critical bugs; maintainable codebase

UX/UI Designer	Minimalist design implementation	User satisfaction score >4.2/5; task completion rate >85%
System Administrator	Infrastructure management, monitoring	99.9% uptime; automated alerting; <1% false positive rate
Security Auditor	Compliance verification	Zero high-severity vulnerabilities; pass security audit
End Users	Productivity improvement	Task completion increase; daily active usage

6. Data Entities & Relationships

6.1 Entity-Relationship Diagram (Conceptual)





Relationship Cardinalities: - **User** → **Task**: One-to-Many (One user creates many tasks) - **User** → **Notification**: One-to-Many (One user receives many notifications) - **Task** → **Tag**: Many-to-Many (Tasks have multiple tags; tags belong to multiple tasks) - **Task_Tag_Junction**: Resolves Many-to-Many relationship with foreign keys

6.2 Data Dictionary

Entity: User

Attribute	Data Type	Constraints	Description
<code>user_id</code>	UUID	PK, Not Null	Unique identifier for each user
<code>email</code>	VARCHAR(255)	Unique, Not Null	User's email address (authentication)
<code>password_hash</code>	VARCHAR(255)	Not Null	Bcrypt hashed password

full_name	VARCHAR(100)	Nullable	User's display name
created_at	TIMESTAMP	Not Null, Default NOW()	Account creation timestamp
last_login	TIMESTAMP	Nullable	Last successful login timestamp
email_verified	BOOLEAN	Not Null, Default FALSE	Email verification status
preferences	JSONB	Nullable	JSON object for user settings (theme, notifications)

Entity: Task

Attribute	Data Type	Constraints	Description
task_id	UUID	PK, Not Null	Unique task identifier
user_id	UUID	FK → User, Not Null	Owning user reference
title	VARCHAR(200)	Not Null	Task title/summary
description	TEXT	Nullable	Detailed task description (rich text)

status	ENUM	Not Null, Default 'pending'	Task status: pending, in_progress, completed
priority	ENUM	Not Null, Default 'low'	Priority level: low, medium, high
due_date	TIMESTAMP	Nullable	Task deadline datetime
created_at	TIMESTAMP	Not Null, Default NOW()	Task creation timestamp
updated_at	TIMESTAMP	Not Null, Default NOW()	Last modification timestamp
completed_at	TIMESTAMP	Nullable	Completion timestamp (when status=completed)
reminder_sent	BOOLEAN	Not Null, Default FALSE	Tracks if reminder was sent

Entity: Tag

Attribute	Data Type	Constraints	Description
tag_id	UUID	PK, Not Null	Unique tag identifier
user_id	UUID	FK → User, Not Null	Tag owner reference

name	VARCHAR(50)	Not Null, Unique per user	Tag name/label
color_hex	CHAR(7)	Not Null, Default '#6B7280'	Color code for visual identification
created_at	TIMESTAMP	Not Null, Default NOW()	Tag creation timestamp
is_archived	BOOLEAN	Not Null, Default FALSE	Soft delete flag

Entity: Notification

Attribute	Data Type	Constraints	Description
notification_id	UUID	PK, Not Null	Unique notification identifier
user_id	UUID	FK → User, Not Null	Recipient user reference
task_id	UUID	FK → Task, Not Null	Associated task reference
type	ENUM	Not Null	Channel: 'push' or 'email'
scheduled_for	TIMESTAMP	Not Null	Planned delivery datetime

<code>delivered_at</code>	TIMESTAMP	Nullable	Actual delivery timestamp
<code>status</code>	ENUM	Not Null, Default 'pending'	Delivery status: pending, sent, failed
<code>retry_count</code>	SMALLINT	Not Null, Default 0	Number of delivery attempts

Junction Table: Task_Tag_Junction

Attribute	Data Type	Constraints	Description
<code>task_id</code>	UUID	PK, FK → Task	Task reference
<code>tag_id</code>	UUID	PK, FK → Tag	Tag reference
<code>created_at</code>	TIMESTAMP	Not Null, Default NOW()	Association timestamp

7. Constraints & Assumptions

7.1 Technical Constraints

ID	Constraint	Impact	Mitigation Strategy
TC-01	Must use MongoDB for primary data storage	Limits relational modeling capabilities	Use embedded documents for 1:1 relationships; implement application-level joins
TC-02	JWT tokens must protect all API endpoints	Requires authentication middleware on every route	Implement centralized auth guard